Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

GO!GO!GO!

Selected Domain

Domain 2 - SME automation for the Orbyte ecosystem

Selected SDG Track (1–17)

SDG 8 - Decent Work and Economic Growth

Project / Solution Name

ChaseLoop - an AI tail-payment chaser for Malaysian SMEs

Name (Leader)

PENG YUERAN

Email & Phone (Leader)

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University Affiliation(s)

Xiamen University Malaysia

Number of Team Members

5

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian small businesses are rarely paid in one go. A renovation, printing or event order is settled with a deposit and a final balance - the tail payment - released only after the work is delivered. That last 30 to 50 per cent is the most contested money in the deal: the customer already has the goods, the supplier holds an unpaid invoice, and the only enforcement mechanism is somebody remembering to ask.

That somebody is the owner. Chasing happens on WhatsApp, between other jobs, from a personal phone. Follow-ups are forgotten in busy weeks, sent too late, or sent too hard and damage a customer worth keeping. Because the chase lives in a chat thread, the owner also cannot answer a simple question: how much of my money is promised but not yet received?

The cost is cash flow. MSMEs contribute 39.1 per cent of Malaysia's GDP and 48.5 per cent of its employment (DOSM, MSME Performance 2023), yet most run receivables on a notebook and a phone. In 12 owner interviews we ran for this project, every owner described the same three burdens: writing a polite reminder in the customer's language, remembering when the next one is due, and judging when "they said they would pay" has become a real problem. Each is small. Together they cost hours weekly and push collection out by weeks.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ChaseLoop is an AI collections co-worker for Malaysian SMEs. It runs the tail-payment chase by itself and hands every decision about money to a human.

Invoices enter by manual entry on a web form, or by automated import: a CSV exported from the company's accounting tool, or a forwarded invoice email the LLM parses. ChaseLoop stores the customer record - name, WhatsApp number, preferred language, invoice number, amount, due date - and schedules the chase: the first reminder one day before the due date, then a follow-up every three days while the balance stays unpaid. Nobody has to approve a send. Sending is the part we automate.

Each reminder is a WhatsApp interactive message with three buttons, the three things a customer can mean: 1. "I have paid" - the customer must attach a payment screenshot before the case can move. 2. "I need more time" - the owner agrees a new date. 3. "There is a problem" - a dispute over amount, quality or delivery.

Option 1 opens a verification task; options 2 and 3 open a negotiation. The AI leaves nothing for the owner to rewrite: it classifies the free-text reply, extracts any promised date and amount, reads the screenshot for amount, date and reference, and shows a one-line summary with a suggested next action. The owner confirms the money actually landed; the ledger then updates itself.

If a customer says nothing for 14 days after the first reminder, the case escalates into the owner's to-do list. Both that window and what happens afterwards are settings, not hard-coded rules.

The SDC Phase 1 workshops map onto this directly: GenAI API calls for drafting and intent extraction, relational schema design for the ledger and message log, and cloud deployment so the scheduler runs without anyone's laptop being open.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Runtime and hosting
Python 3.12, FastAPI, Jinja2 + HTMX owner dashboard, APScheduler for the D-1 reminder and the three-day follow-up. Runs in a Replit Repl today and moves to Replit Deployments before judging; configuration in Replit Secrets.

Data
PostgreSQL (Replit, Neon-backed) for clients, invoices, message log, reply intents and audit trail; SQLite for local development and the test suite.

Messaging
WhatsApp Cloud API (Meta): interactive reply buttons - the API allows a maximum of three, which is exactly our three options - plus media download for payment screenshots and webhooks for delivery, read and reply events.

AI
DeepSeek API (deepseek-chat) for drafting in English, Bahasa Melayu and Mandarin, reply intent classification, date and amount extraction, and reading payment-screenshot fields. No GPU, no self-hosted weights: one HTTPS call per message.

Operations
GitHub for source and review, pytest suite for the scheduler and the intent rules, JSON activity log of every automated send and every owner action.

Block diagram
Owner dashboard and customer WhatsApp -> WhatsApp Cloud API -> FastAPI on Replit (webhook + scheduler + rules) -> PostgreSQL ledger; drafting and classification both call the LLM API. Full diagram in Appendix A.

Domain alignment
Phase 1 keeps the invoice feed pluggable (form, accounting CSV, or the Orbyte hub in Phase 2), so the same chase can run inside the domain ecosystem without depending on it.

Sponsor resources
None of the official SDC sponsor GenAI APIs or cloud credits are used in Phase 1: the prototype runs on Replit with a commercial LLM API billed to the team.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Owners do not distrust automating the sending. They distrust automating the money. Twelve interviews and three shadowing sessions in Klang Valley service SMEs produced that one line, and the whole design follows from it: the system sends, the human decides.

We tested the three-button message on WhatsApp before writing backend code, because the channel set the constraint - Meta allows at most three reply buttons, which forced a messy negotiation down to three intents. Paper prototypes and a clickable Figma flow with the owners produced the 14-day escalation window and the "screenshot or it did not happen" rule.

We kept the build deployable rather than demo-able: PostgreSQL instead of an in-memory store, a real webhook instead of a simulated reply, secrets in Replit Secrets, and a scheduler that survives a restart. Phase 2 adds bank-statement matching and a Bahasa Melayu review of the templates with two pilot companies.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.3 - Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalisation and growth of micro-, small- and medium-sized enterprises. Secondary: SDG 8.2 - achieve higher levels of economic productivity through technological upgrading and innovation, and SDG 9.3 - increase the access of small-scale enterprises to financial services.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In our 12 owner interviews, 9 of 12 chased tail payments manually over WhatsApp, with a median 21 extra days between the agreed due date and the money actually arriving. None could state, at any moment, how much was outstanding and promised. MSMEs are 39.1 per cent of Malaysian GDP (DOSM, 2023), yet there is no published measure of tail-payment delay, which is why the baseline is set from our own interviews. Baseline today: 21 median days late, 3.5 manual follow-ups per invoice, 0 per cent of cases with a timestamped audit trail.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary: median days from due date to confirmed payment. Secondary: share of invoices whose first reminder goes out on D-1 (target 100 per cent); manual follow-ups per invoice; share of customer replies triaged with a one-line summary before the owner opens the case.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Cut median overdue days from 21 to 14 (-33 per cent) and manual follow-ups from 3.5 to 1 per invoice (-70 per cent) across the pilot book; 100 per cent of reminders sent on schedule; 90 per cent of replies summarised and routed within one minute. Measured on the two pilot companies' own ledgers over eight weeks.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

None.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI 0.115 + Uvicorn	code	MIT
APScheduler 3.10	code	MIT
Jinja2 3.1 + HTMX 2.0	code	BSD-3-Clause / MIT
httpx 0.27 + psycopg 3.2	code	BSD-3-Clause / LGPL-3.0
deepseek-chat model	model	Commercial API terms

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

None. No team member is bound by an employment, internship or consulting agreement that could claim ownership of this work.

7. DECLARATION

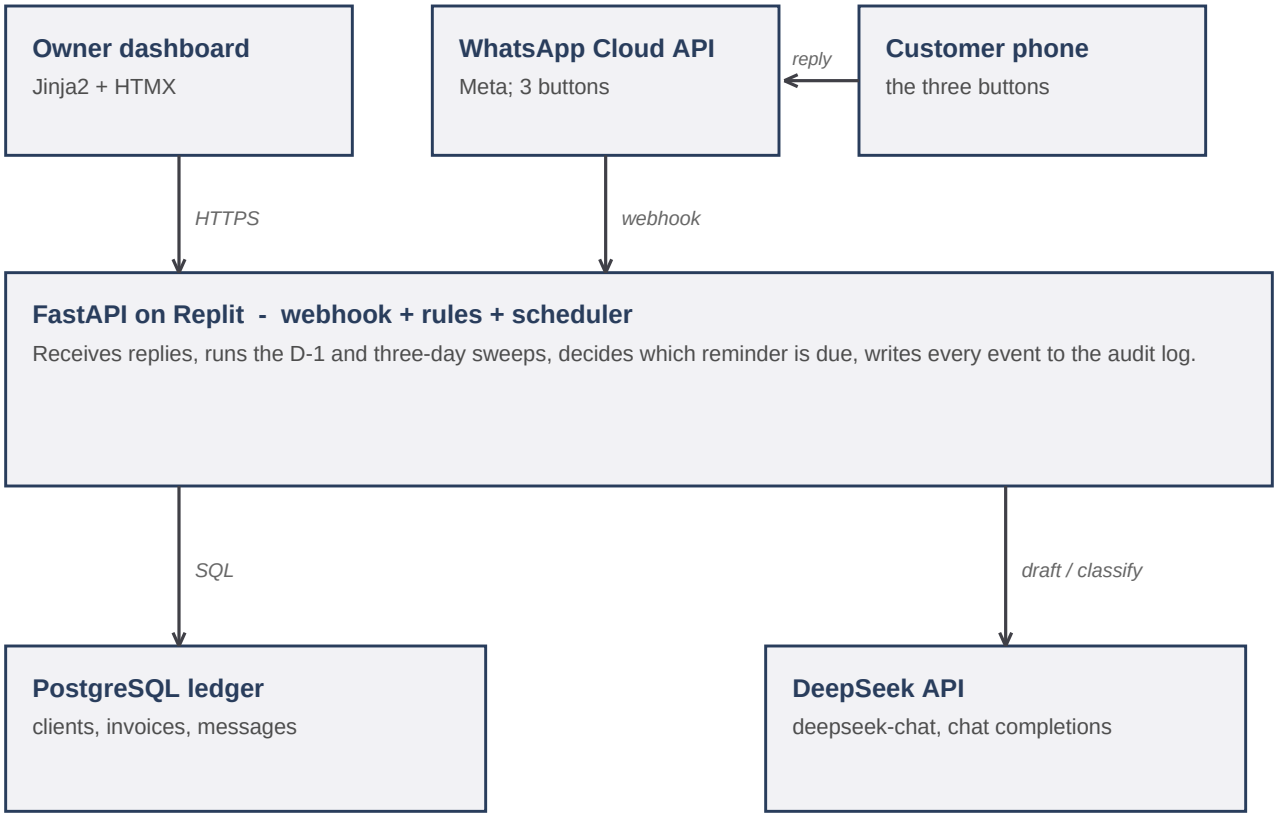
By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

PENG YUERAN

24 October 2026

Appendix A - ChaseLoop system block diagram

Solid boxes are components, arrows are data flows. The system sends reminders on its own; every decision that touches money returns to the owner.



Sending is automatic. Money is not.

The scheduler above runs without any approval step: on D-1 and then every three days the reminder leaves the system by itself. Everything a customer sends back lands in the owner's queue - "I have paid" with a screenshot, a request for more time, or a problem - and no case closes until the owner has verified the money. A customer who stays silent for 14 days after the first reminder is escalated to the owner automatically, and both the window and the escalation path are settings the owner controls.